

Test Model on Multiple Datasets

This notebook evaluates a trained model on three test datasets:

- **Natural:** Uniformly random samples from $[1, 10^{13}]$
- **Cheat:** Numbers with prime factors only within the first 100 primes
- **Non-cheat:** Numbers with at least one prime factor outside the first 100 primes

It reports per-class performance for each dataset.

Configuration

Specify the model checkpoint and encoding to test.

IMPORTANT:

- `ENCODING` should be the **base encoding ONLY** (e.g., `interCRT100`, `CRT100`)
- Do **NOT** include dataset type suffixes like `_natural`, `_cheat`, or `_non_cheat`
- The notebook will automatically test on all three dataset types

Device detection: GPU not available

Using device: cpu

Testing model: `../models/model_interCRT_sqfree_natural/mu/1/best-valid_arithmetic_acc.pth`

Encoding: `interCRT_sqfree`

Task: `mu`

Results will be saved to: `../test_results/interCRT_sqfree_mu`

Helper Functions

Helper functions loaded!

Check if Model Checkpoint Exists

✓ Checkpoint found: `../models/model_interCRT_sqfree_natural/mu/1/best-valid_arithmetic_acc.pth`

Check Test Data Files

Checking test data files:

```
✓ natural      : ../input/input_dir_interCRT_sqfree_natural/mu_interCRT_sqfree_natural.txt.test (106.21 MB)
✓ cheat        : ../input/input_dir_interCRT_sqfree_cheat/mu_interCRT_sqfree_cheat.txt.test (105.97 MB)
✓ non_cheat    : ../input/input_dir_interCRT_sqfree_non_cheat/mu_interCRT_sqfree_non_cheat.txt.test (106.20 MB)
```

Run Evaluation on Each Test Dataset

Evaluation function ready!

Evaluate on Natural Dataset

```
=====
EVALUATING ON NATURAL DATASET
=====
```

Running in CPU mode

Running command:

```
python ../Int2Int/train.py --eval_only True --reload_model /Users/clairexu/Desktop/GitHub/mobius_case_study/notebooks/./models/model_interCRT_sqfree_natural/mu/1/best-valid_arithmetic_acc.pth --eval_data /Users/clairexu/Desktop/GitHub/mobius_case_study/notebooks/./input/input_dir_interCRT_sqfree_natural/mu_interCRT_sqfree_natural.txt.test --eval_size 10000 --data_types int[201]:range(-1,2) --operation data --num_workers...
```

```
=====
Full output saved to: ../test_results/interCRT_sqfree_mu/eval_natural.log
```

✓ Evaluation completed successfully!

Overall Accuracy: 69.97%

Evaluate on Cheat Dataset

EVALUATING ON CHEAT DATASET

Running in CPU mode

Running command:

```
python ../Int2Int/train.py --eval_only True --reload_model /Users/clairexu/Desktop/GitHub/mobius_case_study/notebooks/./models/model_interCRT_sqfree_natural/mu/1/best-valid_arithmetic_acc.pth --eval_data /Users/clairexu/Desktop/GitHub/mobius_case_study/notebooks/./input/input_dir_interCRT_sqfree_cheat/mu_interCRT_sqfree_cheat.txt.test --eval_size 10000 --data_types int[201]:range(-1,2) --operation data --num_workers...
```

Full output saved to: ../test_results/interCRT_sqfree_mu/eval_cheat.log

✓ Evaluation completed successfully!

Overall Accuracy: 88.79%

Evaluate on Non-Cheat Dataset

EVALUATING ON NON-CHEAT DATASET

Running in CPU mode

Running command:

```
python ../Int2Int/train.py --eval_only True --reload_model /Users/clairexu/Desktop/GitHub/mobius_case_study/notebooks/./models/model_interCRT_sqfree_natural/mu/1/best-valid_arithmetic_acc.pth --eval_data /Users/clairexu/Desktop/GitHub/mobius_case_study/notebooks/./input/input_dir_interCRT_sqfree_non_cheat/mu_interCRT_sqfree_non_cheat.txt.test --eval_size 10000 --data_types int[201]:range(-1,2) --operation data --num_workers...
```

Full output saved to: ../test_results/interCRT_sqfree_mu/eval_non_cheat.log

✓ Evaluation completed successfully!

Overall Accuracy: 69.99%

Compile Results

✓ Successfully evaluated on 3 dataset(s)

Results saved to: ../test_results/interCRT_sqfree_mu/all_results.json

Overall Performance Comparison

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OVERALL PERFORMANCE SUMMARY

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	Dataset	Accuracy (%)	Acc $\mu=0$ (%)	Acc $\mu=1$ (%)	Acc $\mu=-1$ (%)	Perfect (%)	Correct (%)	XE Loss
0	Natural	69.97	100.0	100.0	0.0	69.97	69.97	0.438123
1	Cheat	88.79	100.0	100.0	0.0	88.79	88.79	0.151609
2	Non-Cheat	69.99	100.0	100.0	0.0	69.99	69.99	0.431395

Summary saved to: ../test_results/interCRT_sqfree_mu/overall_summary.csv

Precision, Recall, F1, and Confusion Matrix

Note: The Int2Int evaluator provides per-class accuracy but not the full confusion matrix. To compute precision, recall, and F1 properly, we need to run a custom evaluation that captures predictions vs ground truth.

The following section runs a custom evaluation to extract predictions and compute these metrics.

Classification metrics and parsing functions loaded!

Compute Classification Metrics from Evaluation Results

Use the predictions captured from the evaluation runs above:

=====
PARSING PREDICTIONS: NATURAL
=====

Parsing predictions from: ../test_results/interCRT_sqfree_mu/eval_dump/natural/debug/4c3868xexr/eval.valid.arithmetic.0
Extracted 10000 predictions
Metrics computed successfully!

=====
PARSING PREDICTIONS: CHEAT
=====

Parsing predictions from: ../test_results/interCRT_sqfree_mu/eval_dump/cheat/debug/d8wdsdklfe/eval.valid.arithmetic.0
Extracted 10000 predictions
Metrics computed successfully!

=====
PARSING PREDICTIONS: NON_CHEAT
=====

Parsing predictions from: ../test_results/interCRT_sqfree_mu/eval_dump/non_cheat/debug/oqlsf3ugac/eval.valid.arithmetic.0
Extracted 10000 predictions
Metrics computed successfully!

=====
Classification metrics computed for 3 dataset(s)
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Classification Report: Precision, Recall, F1

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CLASSIFICATION REPORT: NATURAL

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Samples evaluated: 10000

Invalid predictions: 0

	precision	recall	f1-score	support
$\mu=-1$	0.00	0.00	0.00	3003
$\mu=0$	1.00	1.00	1.00	3870
$\mu=1$	0.51	1.00	0.68	3127
accuracy			0.70	10000
macro avg	0.50	0.67	0.56	10000
weighted avg	0.55	0.70	0.60	10000

Summary Metrics:

Metric	Macro	Weighted
Precision	0.5034	0.5465
Recall	0.6667	0.6997
F1-Score	0.5585	0.5983

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CLASSIFICATION REPORT: CHEAT

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Samples evaluated: 10000

Invalid predictions: 0

	precision	recall	f1-score	support
$\mu=-1$	0.00	0.00	0.00	1121
$\mu=0$	1.00	1.00	1.00	7962
$\mu=1$	0.45	1.00	0.62	917
accuracy			0.89	10000
macro avg	0.48	0.67	0.54	10000
weighted avg	0.84	0.89	0.85	10000

Summary Metrics:

Metric	Macro	Weighted
Precision	0.4833	0.8375
Recall	0.6667	0.8879
F1-Score	0.5402	0.8531

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CLASSIFICATION REPORT: NON_CHEAT

Samples evaluated: 10000
Invalid predictions: 0

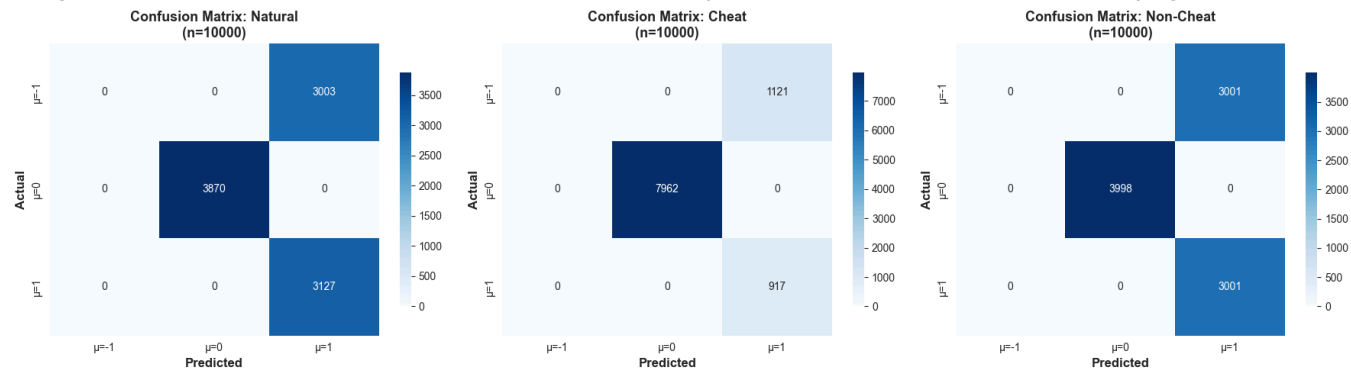
	precision	recall	f1-score	support
$\mu=-1$	0.00	0.00	0.00	3001
$\mu=0$	1.00	1.00	1.00	3998
$\mu=1$	0.50	1.00	0.67	3001
accuracy			0.70	10000
macro avg	0.50	0.67	0.56	10000
weighted avg	0.55	0.70	0.60	10000

Summary Metrics:

Metric	Macro	Weighted
Precision	0.5000	0.5498
Recall	0.6667	0.6999
F1-Score	0.5556	0.5999

Confusion Matrices

Figure saved to: ../test_results/interCRT_sqfree_mu/confusion_matrices.png



Per-Class Precision, Recall, F1

PER-CLASS METRICS: NATURAL

	Class	Precision	Recall	F1-Score
0	$\mu=-1$	0.0000	0.0	0.0000
1	$\mu=0$	1.0000	1.0	1.0000
2	$\mu=1$	0.5101	1.0	0.6756

Saved to: ../test_results/interCRT_sqfree_mu/per_class_metrics_natural.csv

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PER-CLASS METRICS: CHEAT

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	Class	Precision	Recall	F1-Score
0	$\mu=-1$	0.00	0.0	0.0000
1	$\mu=0$	1.00	1.0	1.0000
2	$\mu=1$	0.45	1.0	0.6206

Saved to: ../test_results/interCRT_sqfree_mu/per_class_metrics_cheat.csv

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PER-CLASS METRICS: NON_CHEAT

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	Class	Precision	Recall	F1-Score
0	$\mu=-1$	0.0	0.0	0.0000
1	$\mu=0$	1.0	1.0	1.0000
2	$\mu=1$	0.5	1.0	0.6667

Saved to: ../test_results/interCRT_sqfree_mu/per_class_metrics_non_cheat.csv

Save All Detailed Metrics

All detailed metrics saved to: ../test_results/interCRT_sqfree_mu/detailed_classification_metrics.json

Predictions saved to: ../test_results/interCRT_sqfree_mu/predictions_natural.npz

Predictions saved to: ../test_results/interCRT_sqfree_mu/predictions_cheat.npz

Predictions saved to: ../test_results/interCRT_sqfree_mu/predictions_non_cheat.npz

Manual Inspection: Sample Predictions from Each Test Set

Randomly select 20 samples from each test dataset and display the number, true Mobius value, and model prediction.

natural: loaded 10000 numbers

cheat: loaded 10000 numbers

non_cheat: loaded 10000 numbers

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SAMPLE PREDICTIONS FROM EACH TEST SET

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DATASET: NATURAL (showing 20 random samples)

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Correct: 19/20 (95.0%)

	n	True $\mu(n)$	Predicted $\mu(n)$	Correct
0	(sample 1111)	1	1	Yes
1	(sample 1571)	-1	1	No
2	(sample 1717)	0	0	Yes
3	(sample 2968)	0	0	Yes
4	(sample 3505)	0	0	Yes
5	(sample 3688)	0	0	Yes
6	(sample 4047)	1	1	Yes
7	(sample 4051)	0	0	Yes
8	(sample 4100)	0	0	Yes
9	(sample 4311)	0	0	Yes
10	(sample 4677)	0	0	Yes
11	(sample 6209)	0	0	Yes
12	(sample 7144)	1	1	Yes
13	(sample 7198)	1	1	Yes
14	(sample 8000)	0	0	Yes
15	(sample 8053)	1	1	Yes
16	(sample 9243)	1	1	Yes
17	(sample 9652)	1	1	Yes
18	(sample 9688)	1	1	Yes
19	(sample 9797)	1	1	Yes

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DATASET: CHEAT (showing 20 random samples)

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Correct: 19/20 (95.0%)

	n	True $\mu(n)$	Predicted $\mu(n)$	Correct
0	(sample 1067)	0	0	Yes
1	(sample 1480)	0	0	Yes
2	(sample 2084)	0	0	Yes
3	(sample 2514)	0	0	Yes
4	(sample 2866)	0	0	Yes
5	(sample 2980)	-1	1	No
6	(sample 3371)	1	1	Yes
7	(sample 3393)	0	0	Yes
8	(sample 4253)	0	0	Yes
9	(sample 4421)	0	0	Yes
10	(sample 4837)	0	0	Yes
11	(sample 4936)	0	0	Yes
12	(sample 5423)	0	0	Yes
13	(sample 5515)	0	0	Yes
14	(sample 6384)	0	0	Yes
15	(sample 6531)	0	0	Yes
16	(sample 7539)	1	1	Yes
17	(sample 8264)	0	0	Yes
18	(sample 9527)	0	0	Yes
19	(sample 9777)	0	0	Yes

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DATASET: NON_CHEAT (showing 20 random samples)

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Correct: 16/20 (80.0%)

	n	True $\mu(n)$	Predicted $\mu(n)$	Correct
0	(sample 870)	1	1	Yes
1	(sample 1524)	0	0	Yes
2	(sample 2339)	-1	1	No
3	(sample 3109)	0	0	Yes
4	(sample 3117)	0	0	Yes
5	(sample 3380)	1	1	Yes
6	(sample 3843)	0	0	Yes
7	(sample 4535)	1	1	Yes
8	(sample 4884)	1	1	Yes
9	(sample 4893)	0	0	Yes
10	(sample 5442)	1	1	Yes
11	(sample 5594)	-1	1	No
12	(sample 8149)	1	1	Yes
13	(sample 8378)	0	0	Yes
14	(sample 8665)	-1	1	No
15	(sample 8676)	1	1	Yes
16	(sample 9067)	1	1	Yes
17	(sample 9605)	0	0	Yes
18	(sample 9931)	0	0	Yes
19	(sample 9978)	-1	1	No

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Square-Free Analysis

Analyze the distribution of Möbius function values based on whether numbers are square-free or not.

Note: This analysis is only available for encodings that include the square-free flag (e.g., `n_interCRT_sqfree_mobius`).

Parsing natural test file...
✓ Detected sqfree flag in encoding – using encoded flags.

Total samples: 10000
Square-free: 6130 (61.3%)
Non-square-free: 3870 (38.7%)

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SQUARE-FREE NUMBERS ($\mu(n) \in \{-1, 1\}$)

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True distribution:

$\mu=-1$: 3003 (48.99%)
 $\mu=0$: 0 (0.00%)
 $\mu=1$: 3127 (51.01%)

Predicted distribution:

$\mu=-1$: 0 (0.00%)
 $\mu=0$: 0 (0.00%)
 $\mu=1$: 6130 (100.00%)

Accuracy: 51.01%

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NON-SQUARE-FREE NUMBERS ($\mu(n)$ must be 0)

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True distribution:

$\mu=-1$: 0 (0.00%)
 $\mu=0$: 3870 (100.00%)
 $\mu=1$: 0 (0.00%)

Predicted distribution:

$\mu=-1$: 0 (0.00%)
 $\mu=0$: 3870 (100.00%)
 $\mu=1$: 0 (0.00%)

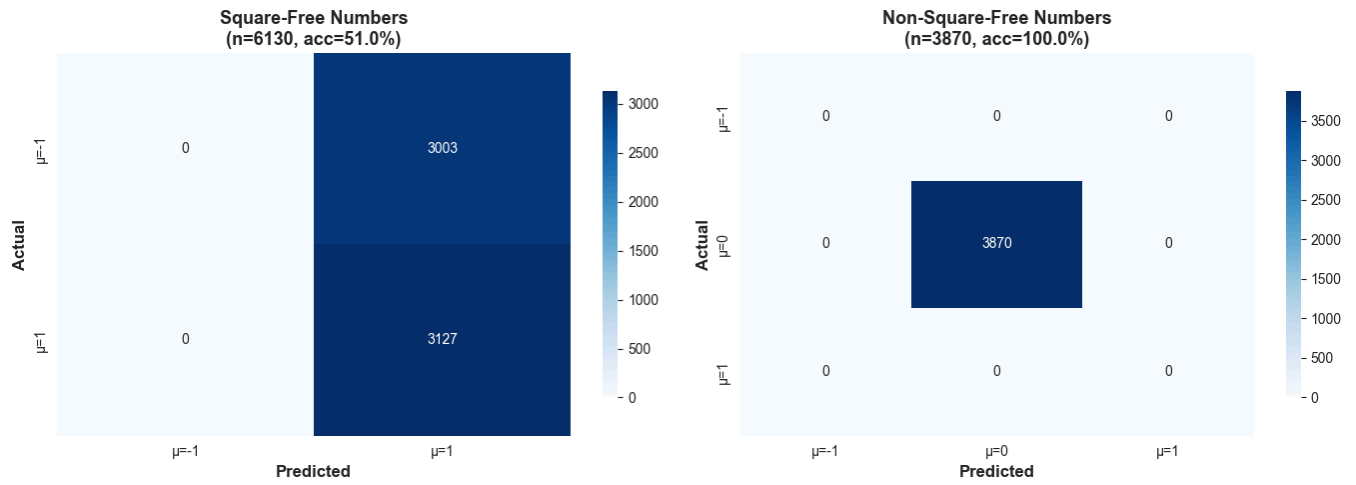
Accuracy: 100.00%

✓ Model correctly predicts $\mu=0$ for all non-square-free numbers!

Square-Free Confusion Matrices

Visualize the confusion matrices separately for square-free and non-square-free numbers.

Figure saved to: ../test_results/interCRT_sqfree_mu/confusion_matrices_square_free.png



CONFUSION MATRIX SUMMARY

Square-free numbers (6130 samples):

Correct predictions: 3127

Accuracy: 51.01%

Non-square-free numbers (3870 samples):

Correct predictions ($\mu=0$): 3870

Accuracy: 100.00%

Violations (predicted $\mu \neq 0$): 0